USER PERSONAS AND EMPATHY MAPS

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine

1. Persona A: Rajesh - Traditional Smallholder Farmer

Rajesh is a 45-year-old farmer who relies on his land for his family's livelihood. He has noticed a decline in yield despite applying more chemical fertilizers year after year.

- Says: 'I want to maximize my yields, but fertilizer and seed costs are becoming unsustainable.'
- Does: Uses traditional crop cycles and applies generic fertilizers based on retail recommendations.
- Thinks: 'Will the rain be sufficient this year? Is my soil losing its fertility permanently?'
- Feels: Anxious about crop failure, market price instability, and lack of technical knowledge.
- Pains: Rising input costs, declining yields, and weather unpredictability.
- Gains: Direct crop suggestions, nutrient corrective guidance, and tools to minimize risk.

? 'I want to maximize yields
but costs are unsustainable.'
? Plants by historical cycles.
? 'Which crops fit my soil?'
? Applies generic fertilizers.
? Manually records harvests.
? Manually records harvests.

? 'Will rain be sufficient
this season?'
? Anxious about crop failures.
? 'Is my soil losing fertility?'
? Stressed about financial loans.
? Optimistic about AI guidance.
? Optimistic about AI guidance.

2. Persona B: Dr. Anita - Agricultural Research Consultant

Dr. Anita conducts research on soil-crop interactions to advise local farmers and agricultural agencies on sustainable management practices.

- Says: 'We need scientific, data-driven frameworks to educate farmers and optimize resource application.'
- Does: Performs soil tests, evaluates crop performance, and compiles technical advisory papers.
- Thinks: 'How can we leverage predictive models to scale agronomic recommendations?'
- Feels: Optimistic about AI in farming but demands transparent and accurate model validation telemetry.
- Pains: Segmented datasets, time-consuming lab processes, and difficulty translating data into simple advice.
- Gains: An analytics dashboard showing model performance comparisons, feature importance, and soil clusters.

Rajesh
(Farmer Persona)